

Sheryl Mathew

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EXPERIENCE

Operations Support Coordinator - Maximus, New York (Remote), United States Sept 2025 – Present

- Engineered **SQL** queries to clean & integrate high-volume OPWDD datasets, improving core data accuracy by **25%**.
- Developed custom **Power BI** dashboards that synthesized statewide expenditure & service utilization trends, automating complex reporting workflows to reduce manual processing time by **40%** for executive stakeholders.
- Analyzed** service utilization patterns across counties, providing data-backed recommendations that optimized resource allocation and service delivery frameworks for over 130k service recipients while improving **operational efficiency**.
- Coordinated with **cross-functional stakeholders** to ensure data integrity, timely outreach, and resource alignment.
- Skills: **SQL, Power BI, Microsoft Excel, Healthcare Operations, Stakeholder Management, HIPAA**

Data Analyst - Deloitte, Hyderabad, India Aug 2023 – Feb 2025

- Led **exploratory data analysis** of large complex financial datasets for **20+** global clients, including multiple **Fortune 500** companies, identifying patterns, detecting fraud transactions, and ensuring data completeness for accurate audits.
- Engineered and automated **35+ ETL pipelines** by migrating legacy **SAS** processes to **Python / PySpark** and **Spark SQL** in **Azure Databricks**, improving performance and increasing audit efficiency by **20%**, saving **1000+** hours annually.
- Designed and developed **10+ interactive dashboards & visualizations** using **Tableau** and **Power BI**, revealing audit trends, risks and financial patterns, and communicated actionable **data insights** with both technical and non-technical stakeholders.
- Developed an **AI-powered RAG** solution for sentiment analysis in ESG reports, reducing manual analysis time by **70%**.
- Mentored junior analysts and received a **Spot Award** for high-quality analysis, attention to detail, and adaptability.
- Skills: **Azure Databricks, Python, PySpark, Spark SQL, SAS, ETL/ELT, Spark DataFrames, Tableau, Power BI, AI / GenAI**

VOLUNTEER EXPERIENCE

Data & Metrics Analyst (Volunteer) - Just for Humanity - New York, United States March 2026 – Present

- Developed reusable **Python** scripts to automate raw data ETL, accelerating reporting cycles and manual data prep time.
- Partnered with leadership to refine **KPIs** and metric definitions to optimize program effectiveness and donor engagement.
- Built and maintained cross-functional **Tableau dashboards** for internal use, making core metrics accessible to stakeholders.

PROJECTS

SmartChart – AI Data Consultant

Python, Streamlit, E2B Sandbox, Hugging Face API, Pandas, Matplotlib, Generative AI, Prompt Engineering 

- Developed a **AI analytics platform** translating natural language into Python code for instant visualization on CSV files.
- Integrated **E2B Code Interpreter** to execute **LLM** generated code in secure sandboxes for complex data manipulation.
- Managed iterative chat history and **prompt engineering** via **Streamlit** and the Leveraged Qwen-2.5-Coder via **Hugging Face API** for code synthesis with iterative chat history management in Streamlit.

Paris Olympics Dashboard

Tableau, Databricks, Python, SQL, Data Analytics, Data Visualization, Business Intelligence  — 

- Implemented an **ETL script** in **Databricks** to ingest and transform Olympics data utilizing joins to create a unified dataset.
- Designed interactive **Tableau dashboards** with sunburst, KPIs, stacked bar, bubble chart, etc. to explore medal distribution.
- Visualized** the impact of athlete migration on medal tallies through filters and drill-downs.

Human Activity Recognition

Python, OpenCV, TensorFlow, Scikit-learn, NumPy, Matplotlib, Machine Learning  — 

- Implemented two **Deep Learning** models for identifying human actions from video data.
- Evaluated **Single-Frame CNN** and **ConvLSTM** architectures, analyzing spatial vs. temporal feature learning.
- Published the work as a **public arXiv preprint**, providing open access to methodology, experiments, and results.

TECHNICAL SKILLS

Languages: Python, SQL **Data:** PySpark, Microsoft Azure Databricks, AWS, Tableau, Power BI, MS Excel, Pandas, NumPy, Matplotlib

Web Dev: HTML, CSS **Tools & AI:** Open AI, Claude, Copilot, LangChain, Prompt Engineering, Hugging Face, GitHub, Git, VS Code

Soft Skills: Written and Verbal Communication, Problem-Solving, Time Management, Presentation, Analytical Skills

EDUCATION

Vellore Institute of Technology, India (2019 – 2023)

Bachelor of Technology - Electronics and Computer Engineering, 8.72/10 CGPA

Coursework: Data Structures and Algorithms, Artificial Intelligence & Machine Learning, Relational Databases

CERTIFICATIONS

- AWS Certified Cloud Practitioner (Validity: Jun 2022 - Jun 2025)
- SQL Analytics on Databricks
- Earned the **SQL 50 badge** on LeetCode by solving 50 MySQL problems covering joins, subqueries, and window functions